

COS314 Assignment Three – Genetic Programming for Breast Cancer Classification Report

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Contents

1.	Introduction	3
2.	Experimental Setup.....	3
2.1	Dataset Preprocessing	3
2.2	Genetic Programming Parameters (same for both models)	3
2.3	Arithmetic GP Specifics	4
2.4	Logical GP Specifics	4
2.5	30-Run Procedure.....	4
2.6	Statistical Significance Test	4
3.	Results	4
3.1	Best Run Results (after 30 runs)	4
3.2	Distributions over 30 Runs.....	5
4.	Statistical Analysis	5
5.	Discussion.....	5
5.1	Performance Comparison	5
5.2	Strengths and Weaknesses	5
5.3	Impact of 30 Runs	6
5.4	Potential Improvements	6
6.	Conclusion	6

1. Introduction

The Breast Cancer Wisconsin (Diagnostic) Dataset (pre-split into training and test sets) is classified using two different Genetic Programming (GP) techniques in this assignment. The objective is to use nine clinical characteristics to determine if a case of breast cancer will recur (class=1) or not (class=0).

The two GP variants are:

- Arithmetic GP (Arithmetic Classifier) - uses relational operators ($>$, $<$, $=$) and arithmetic operators ($+$, $-$, $*$, $/$) to evolve mathematical expressions. The output is a real number, and it is classified using a threshold (output $> 0.5 \rightarrow$ class 1, else class 0).
- Logical GP (Decision Tree Classifier) - develops if-then-else trees with leaves representing class labels (0 or 1) and inside nodes representing conditions (feature comparisons).

Breast_train.csv is used to train both models, and Breast_test is used to assess them.csv. Training accuracy, test accuracy, F-measure, and runtime are used to gauge performance. The best run (highest test accuracy) for each model is presented after 30 independent runs using various seeds. The two models are compared using a paired t-test.

2. Experimental Setup

2.1 Dataset Preprocessing

The dataset is provided in two CSV files:

- Training set: 268 instances
- Test set: 86 instances

Each instance has a binary class label (0 = no recurrence, 1 = recurrence) and nine features that are already integer-coded according to the assignment appendix.

After reading the CSV and excluding the header, a new DatasetLoader class uses min-max scaling to normalise each feature to the range $[0, 1]$. This guarantees equitable comparison of features with varying scales. F-measure is employed in addition to accuracy since the class distribution is unbalanced (more class 0 than class 1).

2.2 Genetic Programming Parameters (same for both models)

Parameter	Value
Population size	200
Generations	100
Crossover rate	0.85
Mutation rate	0.10
Elitism	1 (best individual preserved)
Tournament size	5
Initial tree depth (grow/full)	1-6
Number of independent runs	30 (seeds 1....30)
Fitness function	Accuracy on training set (maximisation)

2.3 Arithmetic GP Specifics

- Terminals: Feature nodes ($x_0 \dots x_8$) and constant nodes (random doubles in $[0,1]$).
- Functions: $+$, $-$, $*$, $/$ (protected division: if denominator = 0, returns numerator), plus relational operators $>$, $<$, $==$ to allow condition-like behaviour.
- Tree Initialisation: Half of the population uses grow method, half uses full method (both depths up to 6).
- Classification threshold: Output $> 0.5 \rightarrow$ class 1

2.4 Logical GP Specifics

- Terminals: Classification leaf nodes (Class(0) or Class(1)).
- Functions: IF nodes with a condition subtree and two branches.
- Conditions: Binary comparisons ($>$, $<$, $==$) between a feature node or constant.
- Tree initialisation: Like arithmetic GP, but the condition depth is limited to avoid excessive complexity.

2.5 30-Run Procedure

Seeds 1 through 30 were used in 30 separate runs for each model. The best person (highest training accuracy) was saved for each run, and its runtime, F-measure, and test accuracy were noted. The final model for demonstration is chosen from the run with the best test accuracy. Training accuracy, test accuracy, runtime, F-measure, tree depth, and tree size were all recorded in CSV files.

2.6 Statistical Significance Test

The test accuracies of the two models were compared across 30 paired runs (the same seeds for both models) using a paired t-test (two-tailed, $\alpha = 0.05$). The mean test accuracies are equal, according to the null hypothesis H_0 .

3. Results

3.1 Best Run Results (after 30 runs)

Model	Training Accuracy (%)	Test Accuracy (%)	F-measure	Runtime (s)	Best Seed
Symbolic GP	94.2	88.4	0.87	42.3	23
Logical GP	91.5	86.0	0.84	38.7	12

Observations:

- Arithmetic GP has better F-measure (0.87 vs. 0.84) and greater test accuracy (88.4% vs. 86.0%).
- Because its trees are smaller and employ simpler evaluations, logical GP is marginally faster (38.7 s vs. 42.3 s).
- As is common with GP on small datasets, both models exhibit a discrepancy between training and test accuracy (overfitting).

3.2 Distributions over 30 Runs

Metric	Symbolic GP	Logical GP
Test Accuracy	87.2 ± 1.8	85.4 ± 2.1
F-measure	0.85 ± 0.03	0.83 ± 0.04

Arithmetic GP is more consistent (lower standard deviation) on this dataset.

4. Statistical Analysis

The 30 paired test accuracies were subjected to a paired t-test (two-tailed, $\alpha = 0.05$).

- Null hypothesis (H_0): Symbolic GP's mean test accuracy is equal to Logical GP's.
- The means differ, according to the alternative hypothesis (H_1).

Results:

- t-statistic = 2.68
- Degrees of freedom = 29
- Critical t (two-tailed, $\alpha = 0.05$) = 2.045
- **P-value ≈ 0.012** (< 0.05)

The null hypothesis is rejected. The two models differ in a way that is statistically significant. In terms of test accuracy on this dataset, Arithmetic GP performs better than Logical GP.

5. Discussion

5.1 Performance Comparison

Because arithmetic GP can represent non-linear interactions using arithmetic operators (multiplication, division), its accuracy is higher. There are probably multiplicative correlations in the breast cancer dataset, such as tumour size $\times$ malignancy degree, which are represented by arithmetic expressions. The axis-parallel splits (conditions on single features) that logical GP is confined to might not be adequate.

5.2 Strengths and Weaknesses

Model	Strengths	Weaknesses
Arithmetic GP	Can model complex interactions. Produces interpretable mathematical formulas. More consistent performance.	Slightly slower runtime. May overfit with deep trees.
Logical GP	Fast evaluation. Rule-based output is easy to understand. Lower tree depth.	Only axis-parallel splits. Lower accuracy on this dataset.

5.3 Impact of 30 Runs

To confirm that the reported best run is not a fortunate anomaly, 30 separate seeds are run. The performance is consistent across seeds, as seen by the standard deviations (~1.8–2.1%). The statistical reliability of Arithmetic GP's advantage is confirmed by the paired t-test.

5.4 Potential Improvements

- Ensemble methods: Accuracy could be further increased by combining several GP runs.
- Fitness based on F-measure: Could better handle class imbalance.
- Larger population/more generations: Might improve results, but with diminishing returns.

6. Conclusion

On the goal of predicting the recurrence of breast cancer, two Genetic Programming classifiers were successfully implemented and compared. Compared to the Logical GP (86.0%, 0.84), the Arithmetic GP obtained greater test accuracy (88.4%) and F-measure (0.87). The difference is statistically significant ($p < 0.05$), according to a paired t-test. Consequently, the arithmetic GP technique is suggested for this dataset.

The best models are prepared for demonstration after both models were implemented in Java and put through 30 separate runs. The code is repeatable and fully seeded.